

Validity Analysis: MILU

Target context: Bangladeshi Bengali University Student — Story Generation Thesis Deployment

Overall risk: **HIGH**

Deployment Context

Use case: The focus is region-dependent generic llm adaptation.

Task: domain-dependent language and dialect understanding

LLMs evaluated: Generic frontier model vs small llms as well as region specific llms

Domain: General llm capability on region-dependent and context driven task understanding

Setting: Evaluating region-dependent llms to assess their capability on local-context tasks.

Target population: A native bengali speaker (a university student doing bsc thesis in story generation) from Dhaka, Bangladesh is using a regional llm to generate local context-driven stories. So it is important to have Bangladesh specific entity presence in the evaluation of the bengali part of the benchmark.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology 	1	Serious concern	high
Input Content 	1	Serious concern	high
Input Form 	4	Minor gaps	high
Output Ontology 	1	Serious concern	high
Output Content 	1	Serious concern	high
Output Form 	1	Serious concern	high
Average	1.5		

 = highest concern  = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MILU is structurally misaligned with the Bangladeshi Bengali story generation deployment along five of six dimensions. Its ontology is India-first by explicit design [Q9, Q19] and excludes Bangladeshi national categories (1971 Liberation War, SSC/HSC, Bangladeshi universities, Ekushe Boi Mela, rural/riverine contexts). Empirical sampling confirms zero Bangladeshi cultural touchstones in 21 Bengali validation items [DATASET-D2-D7], 100% translation rate from English in the sampled Bengali validation split [DATASET-D8] (vs. 25% documented), and Bangladesh appearing only as a distractor option in non-Bengali splits [DATASET-D22, D23]. Ground-truth labels are validated entirely within Indian institutional frameworks [Q8, Q22, Q86] with no Bangladeshi annotators documented. Output form is closed MCQ accuracy [Q51], structurally incompatible with the deployment's open-ended story generation target. The only well-aligned dimension is input form (text in Bangla script [Q29, Q30, DATASET-D24]). High MILU Bengali scores would not predict — and could mislead — the deployment's intended evaluation construct.

ID	Evidence
Q8	"Sshubam Verma1 Mohammed Safi Ur Rahman Khan1,2 Vishwajeet Kumar3 Rudra Murthy3 Jaydeep Sen3 1Nilekani Centre at AI4Bharat 2Indian Institute of Technology, Madras 3IBM Research, India" (p.1)
Q9	"We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams." (p.2)
Q19	"MILU is designed as a culturally relevant benchmark to assess general problem-solving abilities and India-specific knowledge." (p.3)
Q22	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q29	"During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency." (p.4)
Q30	"To remove incorrect language entries, we utilize a combination of INDICLID (Madhani et al., 2023) and Unicode-based filtering (Khan et al., 2024), ensuring that the questions are in the correct language." (p.4)
Q51	"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i))$ " (p.5)
Q86	"We would like to thank EkStep Foundation and Nilekani Philanthropies for their generous grant towards building datasets, models, tools and other resources for Indian languages." (p.9)
D2	ফব্রুয়ারি মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বিধানসভা নির্বাচনে, আম আদম পার্টি (AAP) নমিনলখিতি কোন রাজ্যে সরকার গঠন করছে? ... পাঞ্জাব — Indian state assembly election (AAP in Punjab, 2022) — exclusively India-centric governance content
D3	নির্বাচন কমিশনারকে অপসারণ করা যত্নে পারে ... প্রধান নির্বাচন কমিশনারের সুপারিশে রাষ্ট্রপতি — Removal of India's Election Commissioner — refers to Indian constitutional provisions
D4	'মঙ্গল ভারত' কাজটি নমিনলখিতি কোন জাতীয় নেতার দ্বারা রচিত হয়েছিল ... আচার্য বিনোবা ভাবে — "Mangal Bharat" — Indian national leader Vinoba Bhave; India-specific cultural/literary content
D5	৬৫তম ফিল্মফেয়ার অ্যাওয়ার্ডস, ২০২০-এ সেরা সনিমোটোগ্রাফির জন্য পুরস্কারটি কে জিতেছেন? ... জয় ওজা — 65th Filmfare Awards (Indian film industry awards) — not Bangladeshi film industry
D6	_____ দিল্লিতে কুতুব মিনারের নির্মাণ সম্পন্ন করেছিলেন ... ইলতুতমিশি — Qutub Minar, Delhi — India-specific historical monument
D7	বাহামানি সাম্রাজ্যের প্রথম রাজধানী কী ছিল? ... গুলবার্গা — Bahamani Kingdom capital — medieval Indian/Deccan history, not Bangladeshi
D8	All 21 Bengali examples show is_translated: True — Every Bengali validation example in sample is translated from English
D22	ডিসেম্বর ২০১৯ ৬৫ ভারতীয় ফার্মাকোপোয়া ... বাঙালিদের জন্য — Indian Pharmacopoeia — Bangladesh mentioned as a distractor option, not as subject matter
D23	ডিসেম্বর ২০১৯ ৬৫ ভারতীয় ফার্মাকোপোয়া ... বাঙালিদের জন্য — India Pharmacopoeia question — Bangladesh listed as distractor option2
D24	নচির প্রশ্নে, দেওয়া চারটি বিকল্পের মধ্যে থেকে সঠিক বিকল্পটি নির্বাচন করুন যা প্রদত্ত শব্দের বিপরীত অর্থ প্রকাশ করে। ... শীতল — Antonym question in Bengali — vocabulary test

Practical Guidance

What This Benchmark Measures

MILU measures multiple-choice factual recall on Indian competitive-exam content rendered in Indic scripts, including Bengali. For this deployment, MILU's strongest residual signal is on universal STEM and Bengali grammar/vocabulary items [DATASET-D17-D21, D24-D25] — i.e., baseline academic Bengali competence. It can serve as a screening filter to distinguish

models with zero Bengali competence from those with some, but cannot measure Bangladeshi cultural grounding, register fidelity, or story-generation quality.

Construct Depth

Shallow for the deployment's target construct. MILU probes one cell of a much larger evaluation space (closed-form factual recall on Indian content). The 'culturally relevant' framing in the documentation [Q1, Q19] refers to Indian culture; for Bangladeshi cultural grounding, MILU provides effectively no signal (input_ontology, input_content, output_ontology, output_content all scored 1). The output_form gap (MCQ vs. story generation) further means even strong MILU performance does not validate generative ability.

What Else You Need

Substantial supplementation is required across input_ontology, input_content, output_ontology, output_content, and output_form. Recommended supplements: (1) BLUCK [WEB-15] and BnMMLU [WEB-18] for Bangladesh-sourced cultural and academic MCQ baselines; (2) BengaliMoralBench [WEB-8] and BanglaSocialBench [WEB-16] for Bangladeshi cultural and sociopragmatic context; (3) the Bengali Dialectal Bias Benchmark [WEB-1] for register/dialect coverage; (4) a purpose-built story-generation rubric authored and validated by native Bangladeshi annotators, modeled on BanglaSocialBench's native-speaker-written methodology, covering 1971 Liberation War, post-1971 political history through the 2024 July Uprising, Bangladeshi universities and exam systems, Bangladeshi geography (Dhaka neighborhoods, Padma/Meghna/Jamuna river systems, Cox's Bazar, Chittagong Hill Tracts), Islamic cultural register, and rural/riverine narrative contexts.

ID	Evidence
Q1	"MILU spans 8 domains and 41 subjects across 11 Indic languages, reflecting both general and culturally specific knowledge." (p.1)
Q19	"MILU is designed as a culturally relevant benchmark to assess general problem-solving abilities and India-specific knowledge." (p.3)
D17	স্থির প্রবাহ ট্রান্সফরমার _____ ধরন। ... শলে — Constant current transformer type question — domain-neutral engineering/STEM
D18	নম্নিলখতি কংনগুলি অক্সিডেশন-রডিকশন বক্রিয়া? ... উপরে সবগুলি — Oxidation-reduction reactions — universal science content
D19	হৃদয়ে পসেমকোর দ্বারা চহ্নতি হয়, ... এসএ নোড — SA node as cardiac pacemaker — universal medical content
D20	একটি অ্যাসেম্বলার যা একটি মেশিনে চলে কনিতু অন্য মেশিনে জন্ম মেশিন কোড তরৈকিরে তাকে কবিলা হয়? ... ক্রস-অ্যাসেম্বলার — Cross-assembler definition — universal computer science
D21	দুটি পাত্র আছে X এবং Y। X-এ ১০০ মলি দুধ ... $m = n$ — Milk-water mixture ratio problem — culturally neutral quantitative reasoning
D24	নচিরে প্ৰশ্নে, দেওয়া চারটি বকিল্পরে মধ্য থেকে সই বকিল্পটিনিব্বাচন করুন যা প্রদত্ত শব্দরে বপিরীত অর্থ প্রকাশ করে। ... শীতল — Antonym question in Bengali — vocabulary test
D25	নচিরে বাক্যটির সঠিক সক্রিয় রূপ নিব্বাচন করুন। সবাই তার চমৎকার নৃত্য পরবিশেনা দ্বারা মুগ্ধ হয়েছিল। ... তার চমৎকার নৃত্য পরবিশেনা সবাইকে মুগ্ধ করেছিল। — Active voice conversion of Bengali sentence — grammar in written Bengali
WEB-1	https://arxiv.org/abs/2603.21359
WEB-8	BengaliMoralBench's native Bangladeshi speaker consensus methodology is a model for the annotation approach this deployment requires (https://arxiv.org/abs/2511.03180)
WEB-15	BLUCK uses Bangladesh-sourced MCQ ontology as a category-redesign reference (https://arxiv.org/abs/2505.21092)
WEB-16	BanglaSocialBench's native-speaker-written, context-dependent evaluation methodology is the closest precedent for a story-generation rubric (https://arxiv.org/abs/2603.15949)
WEB-18	BnMMLU sources directly from Bangladeshi NCTB textbooks — viable alternative content provenance (https://arxiv.org/abs/2505.18951)

Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MILU | Context: Bangladeshi Bengali University Student — Story Generation Thesis Deployment

Input Ontology definition: Whether the benchmark’s test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU’s 8-domain, 41-subject taxonomy is explicitly structured around Indian academic and competitive examination content [Q1, Q17, Q19], with ‘local knowledge’ operationalized as Indian state-level [Q12]. The deployment requires Bangladesh-specific categories (1971 Liberation War, SSC/HSC exams, Bangladeshi universities, Ekushe Boi Mela, rural/riverine contexts) which have no designated position in the ontology. Empirical sampling confirms zero Bangladeshi cultural touchstones across 21 Bengali validation items [DATASET-D2, D3, D4, D5, D6, D7] and that Bangladesh appears only as a distractor option in non-Bengali splits [DATASET-D22, D23]. The authors themselves note models perform poorly in culturally relevant areas [Q6, Q68], precisely the domains where Bangladeshi knowledge should appear.

ID	Evidence
Q1	“MILU spans 8 domains and 41 subjects across 11 Indic languages, reflecting both general and culturally specific knowledge.” (p.1)
Q6	“Domain-wise analysis indicates that models perform poorly in culturally relevant areas like Arts & Humanities and Law & Governance compared to general fields like STEM.” (p.1)
Q12	“We focus on region-specific exams to authentically capture local knowledge in the respective language.” (p.2)
Q17	“MILU is a large, multi-domain test set containing multiple-choice based questions (MCQs) taken from over 41 subjects with an emphasis on India-specific knowledge.” (p.3)
Q19	“MILU is designed as a culturally relevant benchmark to assess general problem-solving abilities and India-specific knowledge.” (p.3)
Q68	“Similar trends to those in Section (§5.2) are observed where the open models perform poorly in domains specific to Indian culture—such as Arts & Humanities, Social Sciences, and Law & Governance—but demonstrate higher performance in STEM fields.” (p.7)
D2	ফবেব্রুয়ারি- মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বধিনসভা নরিবাচনে, আম আদমি পার্টি (AAP) নমিনলখিতি ক-োন রাজ্যে সরকার গঠন করছেলি? ... পাঞ্জাব — Indian state assembly election (AAP in Punjab, 2022) — exclusively India-centric governance content
D3	নরিবাচন কমশিনারকে অপসারণ করা যতে পারে ... পুরধান নরিবাচন কমশিনারের সুপারশি রাষ্ট্রপতি — Removal of India’s Election Commissioner — refers to Indian constitutional provisions
D4	‘মঙ্গল ভারত’ কাজটি নমিনলখিতি ক-োন জাতীয় নতোর দ্বারা রচিত হযছেলি ... আচার্য বনি-োবা ভাবে — "Mangal Bharat" — Indian national leader Vinoba Bhave; India-specific cultural/literary content
D5	৬৫তম ফলিমফয়ের অ্যাওয়ার্ডস, ২০২০-এ সরো সনিমোটো-গুয়াফরি জন্য পুরস্কারটি কে জতিছেনে? ... জয় ওজা — 65th Filmfare Awards (Indian film industry awards) — not Bangladeshi film industry
D6	_____ দলিলতিে কুতুব মনিারের নরিমাণ সম্পন্ন করছেলিনে ... ইলতুতমশি — Qutub Minar, Delhi — India-specific historical monument
D7	বাহামান সাম্রাজ্যের পুরথম রাজধানী কী ছিলি? ... গুলবার্গা — Bahamani Kingdom capital — medieval Indian/Deccan history, not Bangladeshi
D22	ডসিমের 2019 ৯৬ ভারতীয় ঔষধাকোষ... বাঙালিদের — Indian Pharmacopoeia — Bangladesh mentioned as a distractor option, not as subject matter
D23	డిసింబర్ 2019లో భారతీయ ఔషధకోషియా ను గుర్తించిన మూడటి దేశం ఏదీ? ... ఆఫ్ఘనిస్తాన్ — India Pharmacopoeia question — Bangladesh listed as distractor option2

Strengths

- Broad multi-domain coverage spans 8 domains including Arts & Humanities, Law & Governance, and STEM [Q1, Q17, DATASET-D2, D4], which means the structural slots exist even if filled with Indian content — a Bangladesh-targeted

re-population of those slots is conceivable.

- Subject-level granularity (41 subjects) and language-stratified evaluation tables [Q99, Q128] provide a fine-grained reporting infrastructure that could in principle host Bangladesh-specific subjects.

ID	Evidence
Q1	"MILU spans 8 domains and 41 subjects across 11 Indic languages, reflecting both general and culturally specific knowledge." (p.1)
Q17	"MILU is a large, multi-domain test set containing multiple-choice based questions (MCQs) taken from over 41 subjects with an emphasis on India-specific knowledge." (p.3)
Q99	"Table 9: Detailed subject level statistics of MILU across different languages." (p.20)
Q128	"Table 47: Detailed subject-wise evaluation for GOOGLE/GEMMA-2-9B-IT on MILU across different languages." (p.52)
D2	ফেব্রুয়ারি- মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বধিনসভা নির্বাচনে, আম আদমি পার্টি (AAP) নমিনলখিতি কনো রাজ্যে সরকার গঠন করছেলি? ... পাঞ্জাব — Indian state assembly election (AAP in Punjab, 2022) — exclusively India-centric governance content
D4	'মঙ্গল ভারত' কাজটিনিমিনলখিতি কনো জাতীয় নতোর দ্বারা রচিত হয়ছেলি ... আচার্য বনোবা ভাবে — "Mangal Bharat" — Indian national leader Vinoba Bhave; India-specific cultural/literary content

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

- IO-1:** Required Bangladesh-specific categories include the 1971 Liberation War, 1952 Language Movement, SSC/HSC exam content, Bangladeshi universities (Dhaka University, BUET, IUT, BRAC), Ekushe Boi Mela and Pohela Boishakh, post-1971 political history including the 2024 July Uprising, rural/riverine contexts, and Bangladeshi literary figures (Humayun Ahmed, Selim Al Deen) [WEB-3, WEB-5, WEB-7]. Story generation also requires categories for narrative settings (rural village life, river delta, Chittagong Hill Tracts, Dhaka neighborhoods) absent from competitive-exam ontology.
- IO-2:** Yes — the taxonomy is structurally India-first [Q9, Q12, Q19] and excludes Bangladeshi national categories. Empirically confirmed: zero Bangladeshi cultural items in the 21-example Bengali validation sample [DATASET-D2 through D7].
- IO-3:** Yes — the taxonomy includes Indian-specific categories (Indian state elections, Indian constitutional law, Filmfare Awards, Indian freedom fighters, Indian state welfare schemes) [DATASET-D2, D3, D5, D9, D10, D11, D12] that are construct-irrelevant or actively misleading for a Bangladeshi cultural-grounding evaluation.
- IO-4:** Major content-validity gap: the taxonomy under-represents Bangladeshi national knowledge and over-represents Indian state/national knowledge. Authors acknowledge poor model performance in culturally relevant areas [Q6, Q68] but attribute it to training data, not to the benchmark's own India-first ontology [Q9].

ID	Evidence
Q6	"Domain-wise analysis indicates that models perform poorly in culturally relevant areas like Arts & Humanities and Law & Governance compared to general fields like STEM." (p.1)
Q9	"We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams." (p.2)
Q12	"We focus on region-specific exams to authentically capture local knowledge in the respective language." (p.2)
Q19	"MILU is designed as a culturally relevant benchmark to assess general problem-solving abilities and India-specific knowledge." (p.3)
Q68	"Similar trends to those in Section (§5.2) are observed where the open models perform poorly in domains specific to Indian culture—such as Arts & Humanities, Social Sciences, and Law & Governance—but demonstrate higher performance in STEM fields." (p.7)

D2	ফেব্রুয়ারি- মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বধিানসভা নব্বিবাচনে, আম আদমি পার্টি (AAP) নমিনলখিতি কোন রাজ্যে সরকার গঠন করছেলি? ... পাঞ্জাব — Indian state assembly election (AAP in Punjab, 2022) — exclusively India-centric governance content
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D5	৬৫তম ফলিমফয়ার অ্যাওয়ার্ডস, ২০২০-এ সেরা সনিমোটোগ্রাফরি জন্য পুরস্কারটিকে জতিছেনে? ... জয় ওজা — 65th Filmfare Awards (Indian film industry awards) — not Bangladeshi film industry
D7	বাহামানি সাম্রাজ্যের প্রথম রাজধানী কী ছিলি? ... গুলবার্গা — Bahamani Kingdom capital — medieval Indian/Deccan history, not Bangladeshi
D9	Which state has topped the India Innovation Index - 2019 published by Niti Aayog? ... Karnataka — India Innovation Index — Indian national governance content
D10	Which one of the following statements about Legislative Committee system of Maharashtra is not correct? — Maharashtra state legislature — India-specific governance at state level
D11	Which state became the first Indian state to launch pension for single women at Rs.1,000/- per month? ... Telangana — Telangana state welfare scheme — Indian state government
D12	Founded in 1950, one of the industrial units owned by Indian Railways is named after the Indian freedom fighter ... Chittaranjan Das — Indian Railways industrial unit named after Indian independence figure
WEB-3	University student literacy supports text-based deployment (https://en.wikipedia.org/wiki/Education_in_Bangladesh)
WEB-5	July 2024 Uprising is a foundational live cultural reference for Bangladeshi students, post-benchmark (https://en.wikipedia.org/wiki/July_Revolution_(Bangladesh))
WEB-7	https://www.chathamhouse.org/2025/11/sheikh-hasina-verdict-and-bangladeshs-upcoming-referendum-signal-transitional-moment-south
WEB-15	BLUCK uses Bangladesh-sourced MCQ ontology as a category-redesign reference (https://arxiv.org/abs/2505.21092)

Information Gaps

- Whether the larger Bengali test split (6,637 items) contains any organically-sourced Bengali-medium items with Bangladeshi-relevant content (validation sample shows 0%)

Requires Expert Verification

- Native Bangladeshi review of any candidate Bengali items that survive translation-filter to confirm absence of Bangladeshi cultural anchoring

Remediation

Gap 1: Taxonomy lacks Bangladeshi-specific categories (1971 Liberation War, SSC/HSC, Bangladeshi universities, cultural events, rural/riverine contexts, 2024 July Uprising)

Recommendation: Supplement with BLUCK (Bangladesh-sourced cultural/historical MCQ across 23 categories [WEB-15]) and BnMMLU (Bangladeshi NCTB-textbook-sourced MMLU [WEB-18]) to populate the missing ontology slots; do not rely on MILU's Bengali split for cultural-knowledge claims.

ID	Evidence
WEB-15	BLUCK uses Bangladesh-sourced MCQ ontology as a category-redesign reference (https://arxiv.org/abs/2505.21092)
WEB-18	BnMMLU sources directly from Bangladeshi NCTB textbooks — viable alternative content provenance (https://arxiv.org/abs/2505.18951)

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MILU | Context: Bangladeshi Bengali University Student — Story Generation Thesis Deployment

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

All MILU content was scraped from Indian competitive exam portals [Q11, Q21, Q88], spanning national civil service, government/private organization, and Indian state-level exams [Q93, Q96, Q97, Q98]. No Bangladeshi exam sources are mentioned. Empirical inspection confirms 100% of sampled Bengali validation items are GPT-4O translations from English [DATASET-D8] (vs. documented 25% [Q43]), and every culturally-specific Bengali item references Indian institutions, persons, or geography [DATASET-D1 through D7]. Identical questions appear verbatim across all 11 language splits [DATASET-D7 cross-config], confirming Bengali content is parallel translation of pan-Indian English exam content rather than Bengali-medium regional sourcing. The deployment's required Bangladeshi register (Cholti with Arabic/Persian loanword density) is not represented; the sampled vocabulary uses Tatsama Sanskritic forms characteristic of pan-Indian written Bengali [DATASET-D1, D25].

ID	Evidence
Q11	"We create this benchmark by collecting questions from over 1500 competitive exams from India." (p.2)
Q21	"We gathered exam-specific questions by scraping various online exam portals that offer previously released question papers from various exams in multiple different languages." (p.3)
Q43	"Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder" (p.4)
Q88	"All data described in this work was scraped from publicly available resources." (p.9)
Q93	"We collected our questions from over 40 exam types ranging from various National and state level civil service examinations to examinations conducted by various government and private organizations." (p.17)
Q96	"Table 6: Overview of various national-level exams and the corresponding years of coverage considered in MILU." (p.18)
Q97	"Table 7: Overview of various government and private organization exams and the corresponding years of coverage considered in MILU." (p.18)
Q98	"Table 8: Overview of various State-level civil services exams and the corresponding years of coverage considered in MILU." (p.18)
D1	১৯৯১ সালে আর্থিক সংকটের কারণ হিসেবে বৈদেশিক মুদ্রার রজি়ার্ড হ্রাসের তাৎক্ষণিক কারণ কী ছিল? ... তলে ও পট্টেরোলিয়াম পণ্যের আমদানির মূল্যেরে তীব্র বৃদ্ধি — India's 1991 balance-of-payments crisis question, translated from English (is_translated=True)
D7	বাহামানি সাম্রাজ্যের প্রথম রাজধানী কী ছিল? ... গুলবার্গা — Bahamani Kingdom capital — medieval Indian/Deccan history, not Bangladeshi
D8	All 21 Bengali examples show is_translated: True — Every Bengali validation example in sample is translated from English
D25	নচিরে বাক্যটির সঠিকি সক্রিয় রূপ নির্বাচন করুন। সবাই তার চমৎকার নৃত্য পরবিশেনা দ্বারা মুগ্ধ হয়েছিল। ... তার চমৎকার নৃত্য পরবিশেনা সবাইকে মুগ্ধ করেছিল। — Active voice conversion of Bengali sentence — grammar in written Bengali

Strengths

- STEM and universal academic content (chemistry, anatomy, computer science, mathematics) is culturally neutral and transfers validly to a Bangladeshi context [DATASET-D17, D18, D19, D20, D21], providing baseline signal on general academic Bengali competence.
- Bengali-language grammar and vocabulary items (antonyms, active/passive transformations) probe written Bengali competence directly [DATASET-D24, D25], offering some linguistic signal even if register-neutral.

ID	Evidence
D17	স্থারি প্রবাহ ট্রান্সফরমার _____ ধরনের। ... শলে — Constant current transformer type question — domain-neutral engineering/STEM
D18	নম্নলিখিত কোনগুলি অক্সিডেশন-রডিকশন বিক্রিয়া? ... উপরে সবগুলি — Oxidation-reduction reactions — universal science content
D19	হৃদয়ের পেসমেকোর দ্বারা চহিনতি হয়, ... এসএ নোড — SA node as cardiac pacemaker — universal medical content
D20	একটি অ্যাসেম্বলার যা একটি মেশিনে চলে কনিতু অন্য মেশিনের জন্য মেশিন কোড তরৈকিরে তাকে কবিলা হয়? ... ক্রস-অ্যাসেম্বলার — Cross-assembler definition — universal computer science
D21	দুটি পাত্র আছে X এবং Y। X-এ ১০০ মলি দুধ ... m = n — Milk-water mixture ratio problem — culturally neutral quantitative reasoning
D24	নচিরে প্রশ্নে, দেওয়া চারটি বিকল্পের মধ্যে থেকে সঠিে বিকল্পটি নিবিাচন করুন যা প্রদত্ত শব্দরে বপিরীত অর্থ প্রকাশ করে। ... শীতল — Antonym question in Bengali — vocabulary test
D25	নচিরে বাক্যটির সঠিক সক্রিয় রূপ নিবিাচন করুন। সবাই তার চমৎকার নৃত্য পরবিশেনা দ্বারা মুগ্ধ হয়েছিল। ... তার চমৎকার নৃত্য পরবিশেনা সবাইকে মুগ্ধ করেছিল। — Active voice conversion of Bengali sentence — grammar in written Bengali

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — the deployment requires Bangladeshi-specific cultural, geographic, and dialectal knowledge (Q1, Q2, Q3 elicitation responses; [WEB-3, WEB-5, WEB-8]). Bangladeshi Cholti register with Arabic/Persian loanwords is required and is distinct from West Bengali norms.

IC-2: No — content reflects Indian national/state cultural framing exclusively [Q9, DATASET-D1 through D7]. The 1991 Indian financial crisis, Indian state elections, Indian constitutional law, Indian film awards, and Indian medieval/colonial monuments all populate the Bengali split.

IC-3: Yes — items requiring Indian-specific knowledge (Indian Election Commissioner removal procedure, AAP state politics, Indian Pharmacopoeia, India's National Food Security Mission) [DATASET-D3, D2, D14] do not transfer and may produce false negatives for users with Bangladeshi civic knowledge.

IC-4: No Bangladeshi annotators participated [Q8, Q86, Q87]; the documentation discloses only Indian institutional affiliation. The flagged_verification_priorities list confirms no Bangladeshi annotators are documented.

IC-5: Severe content-validity violation: Bengali content is (a) translated from English in 100% of sampled items [DATASET-D8], (b) culturally India-centric [DATASET-D2-D7], (c) lacks any Bangladeshi register markers [DATASET-D1, D25]. Strong scores on this content do not predict performance on Bangladesh-grounded story generation.

ID	Evidence
Q1	"MILU spans 8 domains and 41 subjects across 11 Indic languages, reflecting both general and culturally specific knowledge." (p.1)
Q2	"With an India-centric design, MILU incorporates material from regional and state-level examinations, covering topics such as local history, arts, festivals, and laws, alongside standard subjects like science." (p.1)
Q3	"We evaluate over 42 LLMs, and find that current LLMs struggle with MILU, with GPT-4o achieving the highest average accuracy at 74%." (p.1)
Q8	"Sshubam Verma ¹ Mohammed Safi Ur Rahman Khan ^{1,2} Vishwajeet Kumar ³ Rudra Murthy ³ Jaydeep Sen ³ Nilekani Centre at AI4Bharat 2Indian Institute of Technology, Madras 3IBM Research, India" (p.1)
Q9	"We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams." (p.2)

Q40	"For subjects with insufficient questions, we sampled questions from the English set from that subject and translated them into the required language using GPT-4O." (p.4)
Q43	"Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder" (p.4)
Q86	"We would like to thank EkStep Foundation and Nilekani Philanthropies for their generous grant towards building datasets, models, tools and other resources for Indian languages." (p.9)
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
D1	১৯৯১ সালে আর্থিক সংকটের কারণ হিসেবে বদিশে মুদ্রার রজির্ভ হ্রাসের তাৎক্ষণিক কারণ কী ছিল? ... তলে ও পট্টেরোলিয়াম পণ্যের আমদানির মূল্যের তীব্র বৃদ্ধি — India's 1991 balance-of-payments crisis question, translated from English (is_translated=True)
D2	ফব্রুয়ারি- মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বধিনসভা নর্বিচাচনে, আম আদমপারটি (AAP) নর্মিনলখিতি কংন রাজ্যে সরকার গঠন করছেলি? ... পাঞ্জাব — Indian state assembly election (AAP in Punjab, 2022) — exclusively India-centric governance content
D3	নর্বিচাচন কমশিনারকে অপসারণ করা যতে পারে ... প্রধান নর্বিচাচন কমশিনারের সুপারশি রাষ্ট্রপতি — Removal of India's Election Commissioner — refers to Indian constitutional provisions
D4	'মঙ্গল ভারত' কাজটি নর্মিনলখিতি কংন জাতীয় নতোর দ্বারা রচতি হচ্ছেলি ... আচার্য বনিংবা ভাবে — "Mangal Bharat" — Indian national leader Vinoba Bhave; India-specific cultural/literary content
D5	৬৫তম ফলিমফয়ার অ্যাওয়ার্ডস, ২০২০-এ সেরা সনিমোটোগ্রাফরি জন্য পুরস্কারটিকে জতিছেনে? ... জয় ওজা — 65th Filmfare Awards (Indian film industry awards) — not Bangladeshi film industry
D6	_____ দলিলতিে কুতুব মনিারের নর্মাণ সম্পন্ন করছেলিনে ... ইলতুতমশি — Qutub Minar, Delhi — India-specific historical monument
D7	বাহমানি সাম্রাজ্যের প্রথম রাজধানী কী ছিলি? ... গুলবার্গা — Bahamani Kingdom capital — medieval Indian/Deccan history, not Bangladeshi
D8	All 21 Bengali examples show is_translated: True — Every Bengali validation example in sample is translated from English
D14	জাতীয় খাদ্য সুরক্ষা মশিন একটি ফসল উন্নয়ন প্রকল্প ... জাতীয় খাদ্য সুরক্ষা মশিন আগস্ট ২০০৭ সালে চালু হচ্ছেলি — India's National Food Security Mission — Indian government program
D25	নচিরে বাক্যটির সঠিকি সক্রয়ি রূপ নর্বিচাচন করুন। সবাই তার চমৎকার নৃত্য পরবিশেনা দ্বারা মুগ্ধ হচ্ছেলি। ... তার চমৎকার নৃত্য পরবিশেনা সবাইকে মুগ্ধ করছেলি। — Active voice conversion of Bengali sentence — grammar in written Bengali
WEB-3	University student literacy supports text-based deployment (https://en.wikipedia.org/wiki/Education_in_Bangladesh)
WEB-5	July 2024 Uprising is a foundational live cultural reference for Bangladeshi students, post-benchmark (https://en.wikipedia.org/wiki/July_Revolution_(Bangladesh))
WEB-8	BengaliMoralBench's native Bangladeshi speaker consensus methodology is a model for the annotation approach this deployment requires (https://arxiv.org/abs/2511.03180)

Information Gaps

- Whether the test split (6,637 Bengali items) has a different translation rate or contains organically-sourced Bengali-medium items with West-Bengali-specific (but not Bangladeshi) cultural content
- Systematic register classification of Bengali items by a Bangladeshi linguist

Requires Expert Verification

- Native Bangladeshi linguistic audit of register markers across a larger Bengali sample
- Comparison of MILU Bengali content to BLUCK/BnMMLU Bangladeshi-sourced content for cultural-distance measurement

Remediation

Gap 1: 100% of sampled Bengali validation items are GPT-4O translations from English Indian-exam content [DATASET-D8]; no Bangladeshi register markers; no Bangladeshi named entities

Recommendation: Source supplementary content from Bangladesh-origin exams (BCS, SSC, HSC, university entrance) per BLUCK/BnMMLU methodology; for the thesis, document explicitly that MILU Bengali scores reflect translated pan-Indian content and report MILU only as a baseline-screening signal alongside Bangladesh-sourced benchmarks.

Gap 2: Bangladeshi Cholti register (Arabic/Persian loanword density) is absent from the sampled translated Bengali content [DATASET-D1, D25]; West-Bengali/pan-Indian Sanskrit register dominates

Recommendation: Supplement with the Bengali Dialectal Bias Benchmark [WEB-1] for explicit dialect coverage (including Barishal, Chittagong, Sylhet, Rangpur, Noakhali); commission a register-stratified sample of story-generation prompts that probes Bangladeshi Cholti vs. West-Bengali outputs.

ID	Evidence
D1	১৯৯১ সালে আর্থিক সংকটের কারণ হিসেবে বৈদেশী মুদ্রার রজিার্ড হ্রাসের তাৎক্ষণিক কারণ কী ছিল? ... তলে ও পট্রোলিয়াম পণ্যের আমদানির মূল্যের তীব্র বৃদ্ধি— India's 1991 balance-of-payments crisis question, translated from English (is_translated=True)
D8	All 21 Bengali examples show is_translated: True — Every Bengali validation example in sample is translated from English
D25	নচিরে বাক্যটির সঠিক সক্রিয় রূপ নির্বাচন করুন। সবাই তার চমৎকার নৃত্য পরবিশেনা দ্বারা মুগ্ধ হয়েছিল। ... তার চমৎকার নৃত্য পরবিশেনা সবাইকে মুগ্ধ করেছিল। — Active voice conversion of Bengali sentence — grammar in written Bengali
WEB-1	https://arxiv.org/abs/2603.21359

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: MILU | Context: Bangladeshi Bengali University Student — Story Generation Thesis Deployment

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU is purely text-based MCQ in Indic scripts [Q29] with INDICLID + Unicode filtering for script correctness [Q30] and manual verification per language [Q32]. The deployment is also text-only in Bengali script [WEB-2]. Empirical inspection confirms Bengali items are correctly rendered in Bangla script with no encoding errors [DATASET-D24, D25]. No script-level mismatch exists. The remaining concerns (register/dialect stratification, translation-induced surface form) are content/lexical rather than form properties; the writing-system-level form alignment is strong. One residual concern is that 100% translation rate in the Bengali validation sample [DATASET-D8] may produce translation-flavored surface text that differs subtly from organically Bengali-written prose, but this is at the boundary of form vs. content.

ID	Evidence
Q29	“During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency.” (p.4)
Q30	“To remove incorrect language entries, we utilize a combination of INDICLID (Madhani et al., 2023) and Unicode-based filtering (Khan et al., 2024), ensuring that the questions are in the correct language.” (p.4)
Q32	“As a final step, we manually verify a sample of questions from each language to ensure accuracy and correct any remaining errors.” (p.4)
D8	All 21 Bengali examples show is_translated: True — Every Bengali validation example in sample is translated from English
D24	নচিৰে প্ৰশ্নে, দোয়া চাৰটি বকিল্পৰে মধ্য থকোঁ সোঁ বকিল্পটি নিৰ্বাচন কৰুন যা প্ৰদত্ত শব্দৰে বপিত অৰ্থ প্ৰকাশ কৰে। ... শীতল — Antonym question in Bengali — vocabulary test
D25	নচিৰে বাক্যটিৰ সঠিক স্ক্ৰিপ্ট বুলি নিৰ্বাচন কৰুন। সবাই তাৰ চমৎকাৰ নৃত্য পৰিবেশনা দ্বাৰা মুগ্ধ হৈছিল। ... তাৰ চমৎকাৰ নৃত্য পৰিবেশনা সবাইকো মুগ্ধ কৰিছিল। — Active voice conversion of Bengali sentence — grammar in written Bengali
WEB-2	Bengali alphasyllabary tokenization challenges are documented; Sangraha corpus has ~30B Bengali tokens (https://arxiv.org/abs/2507.23248)

Strengths

- Bengali script (Bangla lipi) rendering is correct and well-formed throughout the sampled split [DATASET-D24, D25]; no script contamination from other Indic scripts observed.
- Documentation explicitly applies INDICLID and Unicode-based filtering to ensure script and language correctness [Q30], plus a final manual sample verification per language [Q32].
- Modality (text) matches the deployment's text-only requirement; no image/audio dependency [Q29].

ID	Evidence
Q29	“During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency.” (p.4)
Q30	“To remove incorrect language entries, we utilize a combination of INDICLID (Madhani et al., 2023) and Unicode-based filtering (Khan et al., 2024), ensuring that the questions are in the correct language.” (p.4)
Q32	“As a final step, we manually verify a sample of questions from each language to ensure accuracy and correct any remaining errors.” (p.4)
D24	নচিৰে প্ৰশ্নে, দোয়া চাৰটি বকিল্পৰে মধ্য থকোঁ সোঁ বকিল্পটি নিৰ্বাচন কৰুন যা প্ৰদত্ত শব্দৰে বপিত অৰ্থ প্ৰকাশ কৰে। ... শীতল — Antonym question in Bengali — vocabulary test

D25	নচিৰে বাক্যটৰি সঠকি সক্রয়ি ৰূপ নৰিবাচন কৰুন। সবাই তাৰ চমৎকাৰ নৃত্য পৰবিশেণা দ্বাৰা মুগ্ধ হয়ছেলি। ... তাৰ চমৎকাৰ নৃত্য পৰবিশেণা সবাইকে মুগ্ধ কৰছেলি। — Active voice conversion of Bengali sentence — grammar in written Bengali
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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal modality (written Bengali script) matches deployment [Q29, WEB-2, DATASET-D24]. No resolution/encoding mismatch — both are Unicode Bengali text.

IF-2: Bangladeshi infrastructure supports Bengali text input/output; urban Dhaka university students have adequate connectivity and devices [WEB-9, WEB-10]. No infrastructure barrier to MILU's input form.

IF-3: MILU excludes reading-comprehension and image-based questions for uniformity [Q29], which slightly narrows form coverage relative to general LLM use, but is acceptable for an MCQ knowledge benchmark.

IF-4: Minor concern: cross-language parallel-translation patterns [DATASET-D7] may produce translationese surface forms in Bengali items, distinct from organically-written Bengali prose; this is a subtle form-quality issue rather than a structural mismatch.

ID	Evidence
Q29	"During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency." (p.4)
D7	বাহামান সাম্ৰাজ্যৰে প্ৰথম ৰাজধানী কী ছিলি? ... গুলবার্গা — Bahamani Kingdom capital — medieval Indian/Deccan history, not Bangladeshi
D24	নচিৰে প্ৰশ্ননে, দেওয়া চাৰটিকিল্পৰে মধ্য থেকে সেই বকিল্পটিনিৰিবাচন কৰুন যা প্ৰদত্ত শব্দৰে বপিৰীত অৰ্থ প্ৰকাশ কৰে। ... শীতল — Antonym question in Bengali — vocabulary test
WEB-2	Bengali alphasyllabary tokenization challenges are documented; Sangraha corpus has ~30B Bengali tokens (https://arxiv.org/abs/2507.23248)
WEB-9	Urban Dhaka connectivity supports text-based deployment (https://datareportal.com/reports/digital-2024-bangladesh)
WEB-10	https://www.tbsnews.net/bangladesh/over-50-households-use-internet-bangladesh-bbs-1032411

Information Gaps

- Whether translation-induced surface patterns systematically differ from organically-written Bangladeshi Bengali prose at a level that affects model evaluation

Requires Expert Verification

- Native Bangladeshi reader assessment of whether translated Bengali surface form reads naturally

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MILU | Context: Bangladeshi Bengali University Student — Story Generation Thesis Deployment

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU's output space is a discrete label among up to four MCQ options [Q29, Q51], with the 41-subject taxonomy serving as metadata for analysis [Q38]. The deployment targets open-ended Bengali story generation, where the output space is free-form narrative text — fundamentally incompatible with a closed MCQ label space [DATASET-D5 OF concern]. Furthermore, the correct-answer category encodings reflect Indian national/state factual standards — Indian Election Commissioner removal procedure, Indian state political wins, Indian Pharmacopoeia [DATASET-D2, D3, D14] — which encode non-Bangladeshi values and assumptions in the answer-key structure itself. The MCQ output ontology cannot be re-mapped to evaluate culturally grounded narrative generation, and within-Bengali items the answer correctness is calibrated to Indian rather than Bangladeshi knowledge frameworks. No mechanism distinguishes India-specific from Bangladesh-specific answer correctness.

ID	Evidence
Q29	"During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency." (p.4)
Q38	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
Q51	"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i))$ " (p.5)
D2	ফেব্রুয়ারি- মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বধিানসভা নব্বিবাচনে, আম আদমি পার্টি (AAP) নমিনলখিতি কোন রাজ্যে সরকার গঠন করছেলি? ... পাঞ্জাব — Indian state assembly election (AAP in Punjab, 2022) — exclusively India-centric governance content
D3	নব্বিবাচন কমিশনারকে অপসারণ করা যতে পারে ... প্রধান নব্বিবাচন কমিশনারের সুপারশি রাষ্ট্রপতি — Removal of India's Election Commissioner — refers to Indian constitutional provisions
D5	৬৫তম ফলিমফয়ার অ্যাওয়ার্ডস, ২০২০-এ সরো সনিমোটো গ্ৰাফরি জন্য পুরস্কারটিকে জতিছেনে? ... জয় ওজা — 65th Filmfare Awards (Indian film industry awards) — not Bangladeshi film industry
D14	জাতীয় খাদ্য সুরক্ষা মশিন একট ফিসল উন্নয়ন প্রকল্প ... জাতীয় খাদ্য সুরক্ষা মশিন আগস্ট ২০০৭ সালে চালু হয়েছিল — India's National Food Security Mission — Indian government program

Strengths

- Structured 4-option categorical output is reproducible and unambiguously scorable [Q51], which provides a clean baseline for measuring factual recall on whatever knowledge frame is encoded — useful as a first-pass screen even if not for the deployment's primary task.

ID	Evidence
Q51	"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i))$ " (p.5)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output labels are India-calibrated correct answers [DATASET-D3 — Indian CEC removal procedure; DATASET-D2 — Indian state election winner]. For a Bangladeshi user, the 'correct' answer for governance items presupposes Indian constitutional structures, not Bangladeshi ones.

OO-2: Missing: any output category encoding Bangladeshi-specific correct answers (e.g., Bangladesh's Election Commission structure, SSC/HSC curriculum standards, Bangladeshi historical events). No story-generation-relevant output categories exist at all.

OO-3: Yes — categories encode Indian national values/assumptions: Indian-state-level governance correctness [DATASET-D2, D3], Indian historical canon [DATASET-D6, D7], Indian cultural figures [DATASET-D4, D5]. These violate convergent validity for a Bangladeshi-targeted evaluation.

OO-4: Stakeholder-driven taxonomy redesign would be required: a story-generation rubric grounded in Bangladeshi cultural touchstones, register, and named entities is needed — MCQ accuracy cannot be re-purposed. BanglaSocialBench's native-speaker-written context-dependent methodology [WEB-16] is the closest existing model.

OO-5: Severe structural and content validity violations: the output taxonomy is doubly misaligned (closed-form MCQ vs. open generation; Indian answer-keys vs. Bangladeshi correctness).

ID	Evidence
Q29	"During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency." (p.4)
Q38	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
Q51	"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i))$ " (p.5)
D2	ফব্রুয়ারি- মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বধিনসভা নির্বাচনে, আম আদমি পার্টি (AAP) নমিনলখিতি কণে ন রাজ্যে সরকার গঠন করছেলি? ... পাঞ্জাব — Indian state assembly election (AAP in Punjab, 2022) — exclusively India-centric governance content
D3	নর্বিবাচন কমশিনারকে অপসারণ করা যতে পারে ... প্রধান নর্বিবাচন কমশিনারের সুপারশি রাষ্ট্রপতি — Removal of India's Election Commissioner — refers to Indian constitutional provisions
D4	'মঙ্গল ভারত' কাজটিনমিনলখিতি কণে ন জাতীয় নতোর দ্বারা রচতি হয়ছেলি ... আচার্য বনিবা ভাবে — "Mangal Bharat" — Indian national leader Vinoba Bhave; India-specific cultural/literary content
D5	৬৫তম ফলিমফয়ার অ্যাওয়ার্ডস, ২০২০-এ সরো সনিমোটো:গ্ৰাফরি জন্য পুরস্কারটিকে জতিছেনে? ... জয় ওজা — 65th Filmfare Awards (Indian film industry awards) — not Bangladeshi film industry
D6	_____ দলিলতিে কুতুব মনিারেরে নরিমাণ সম্পন্ করছেলিনে ... ইলতুতমশি — Qutub Minar, Delhi — India-specific historical monument
D7	বাহামানি সাম্রাজ্যেরে প্রথম রাজধানী কী ছিলি? ... গুলবার্গা — Bahamani Kingdom capital — medieval Indian/Deccan history, not Bangladeshi
WEB-15	BLUCK uses Bangladesh-sourced MCQ ontology as a category-redesign reference (https://arxiv.org/abs/2505.21092)
WEB-16	BanglaSocialBench's native-speaker-written, context-dependent evaluation methodology is the closest precedent for a story-generation rubric (https://arxiv.org/abs/2603.15949)

Information Gaps

- Whether any items in the test split have answer keys that would also be correct under Bangladeshi knowledge frameworks (universal STEM items presumably yes; cultural/governance items presumably no)

Requires Expert Verification

- Bangladeshi expert review of cultural/governance answer keys to identify items where Indian and Bangladeshi correct answers diverge

Remediation

Gap 1: Closed MCQ output space cannot evaluate open-ended story generation; correct-answer categories encode Indian rather than Bangladeshi knowledge

Recommendation: Construct a story-generation rubric with native Bangladeshi linguists and literary scholars, modeled on BanglaSocialBench's native-speaker-written context-dependent methodology [WEB-16], covering cultural authenticity, register fidelity, named-entity grounding, and narrative coherence.

ID	Evidence
WEB-16	BanglaSocialBench's native-speaker-written, context-dependent evaluation methodology is the closest precedent for a story-generation rubric (https://arxiv.org/abs/2603.15949)

Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MILU | Context: Bangladeshi Bengali University Student — Story Generation Thesis Deployment

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Ground truth labels were validated by Indian exam portal subject experts [Q22] and audited by AI4Bharat volunteers [Q87]. The benchmark team is exclusively affiliated with Indian institutions — Nilekani Centre at AI4Bharat, IIT Madras, IBM Research India [Q8] — and funded by Indian philanthropies [Q86]. No Bangladeshi reviewers or subject-matter experts are mentioned anywhere. The elicitation explicitly designates Indian-only annotator pools (including West Bengali speakers) as 'explicitly insufficient' for this deployment. Empirically, the cultural/governance items in the Bengali split [DATASET-D2, D3, D4, D5, D6] have answer labels reflecting Indian standards, and no Bangladeshi institutional or individual validator is documented. For governance items like Election Commissioner removal [DATASET-D3], a Bangladeshi annotator would correctly identify the framing as Indian-specific. No formal IAA, annotator demographic profiles, or aggregation methodology disclosures are provided.

ID	Evidence
Q8	"Shubam Verma1 Mohammed Safi Ur Rahman Khan1,2 Vishwajeet Kumar3 Rudra Murthy3 Jaydeep Sen3 1Nilekani Centre at AI4Bharat 2Indian Institute of Technology, Madras 3IBM Research, India" (p.1)
Q22	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q86	"We would like to thank EkStep Foundation and Nilekani Philanthropies for their generous grant towards building datasets, models, tools and other resources for Indian languages." (p.9)
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
D2	ফবেয়ুয়ারি- মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বধিানসভা নির্বাচনে, আম আদমি পার্টি (AAP) নমিনলখিতি কংন রাজ্যে সরকার গঠন করছেলি? ... পাঞ্জাব — Indian state assembly election (AAP in Punjab, 2022) — exclusively India-centric governance content
D3	নির্বাচন কমিশনারকে অপসারণ করা যতে পারে ... প্রধান নির্বাচন কমিশনারের সুপারশি রাষ্ট্রপতি — Removal of India's Election Commissioner — refers to Indian constitutional provisions
D4	'মঙ্গল ভারত' কাজটি নমিনলখিতি কংন জাতীয় নতোর দ্বারা রচিত হয়ছেলি ... আচার্য বনিবা ভাবে — "Mangal Bharat" — Indian national leader Vinoba Bhave; India-specific cultural/literary content
D5	৬৫তম ফলিমফয়ার অ্যাওয়ার্ডস, ২০২০-এ সেরা সনিমোটোগ্রাফরি জন্য পুরস্কারটিকে জতিছেনে? ... জয় ওজা — 65th Filmfare Awards (Indian film industry awards) — not Bangladeshi film industry
D6	_____ দলিলতি কুতুব মনিাররে নির্মাণ সম্পন্ন করছেলি ... ইলতুতমশি — Qutub Minar, Delhi — India-specific historical monument

Strengths

- Source exam portals provided subject-expert answer validation [Q22], establishing baseline label correctness within the Indian-exam knowledge frame.
- Manual cluster review and AI4Bharat volunteer audits [Q37, Q87] add a second pass of human verification, even if not Bangladeshi-grounded.
- Approximately 45% of items arrived with topic tags from source portals [Q33], reducing reliance on automated clustering for those items.

ID	Evidence
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Q22	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q33	"Upon examination, we found that approximately 45% of questions were accurately labeled with a topic name, while the remaining questions lacked this information." (p.4)
Q37	"We manually review these clusters and assign appropriate subject labels." (p.4)
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: No — labels reflect Indian stakeholder perspectives only [Q8, Q22]. The elicitation requires native Bangladeshi Bengali speakers as primary ground truth and explicitly designates Indian-only pools as insufficient.

OC-2: Significant disagreement is likely on cultural/governance items where Indian and Bangladeshi correct answers diverge structurally [DATASET-D3 — Indian-specific CEC removal procedure]. Universal STEM items would show low disagreement.

OC-3: Documentation discloses Indian institutional affiliation (AI4Bharat, IIT Madras, IBM Research India [Q8]) and Indian funding [Q86], but provides no formal Datasheet, demographic breakdown, or inter-annotator agreement statistics.

OC-4: Re-annotation by a representative Bangladeshi annotator pool would be required for any deployment-relevant use; existing labels cannot be assumed valid for Bangladesh-targeted evaluation.

OC-5: No aggregation methodology disclosed; minority-perspective erasure cannot be assessed from documentation. INSUFFICIENT DOCUMENTATION on aggregation procedure.

OC-6: Severe convergent and external validity violations: labels do not correlate with regional Bangladeshi perspectives on cultural and governance items, and Indian judgments will not generalize.

ID	Evidence
Q8	"Sshubam Verma1 Mohammed Safi Ur Rahman Khan1,2 Vishwajeet Kumar3 Rudra Murthy3 Jaydeep Sen3 1Nilekani Centre at AI4Bharat 2Indian Institute of Technology, Madras 3IBM Research, India" (p.1)
Q22	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q86	"We would like to thank EkStep Foundation and Nilekani Philanthropies for their generous grant towards building datasets, models, tools and other resources for Indian languages." (p.9)
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
D3	নব্বিবাচন কমিশনারকে অপসারণ করা যত্নে পারবে ... প্রধান নব্বিবাচন কমিশনারের সুপারিশে রাষ্ট্রপতি — Removal of India's Election Commissioner — refers to Indian constitutional provisions
WEB-8	BengaliMoralBench's native Bangladeshi speaker consensus methodology is a model for the annotation approach this deployment requires (https://arxiv.org/abs/2511.03180)
WEB-16	BanglaSocialBench's native-speaker-written, context-dependent evaluation methodology is the closest precedent for a story-generation rubric (https://arxiv.org/abs/2603.15949)

Information Gaps

- Inter-annotator agreement statistics (not disclosed)
- Annotator demographic breakdown beyond institutional affiliation (not disclosed)
- Aggregation methodology for resolving disagreements (not disclosed)

Requires Expert Verification

- Bangladeshi annotator re-review of culturally/politically specific Bengali items to quantify the Indian-vs-Bangladeshi label divergence rate

Remediation

Gap 1: Ground-truth labels validated only by Indian institutional reviewers [Q8, Q22, Q86]; no Bangladeshi annotators; elicitation declares Indian-only pools 'explicitly insufficient'

Recommendation: Recruit a panel of native Bangladeshi Bengali annotators (university-educated, cross-regional including Dhaka, Sylhet, Chittagong, Barisal) to (a) re-validate any MILU Bengali items used in the thesis, and (b) author ground truth for the new story-generation rubric. Disclose annotator demographics and inter-annotator agreement in the thesis.

ID	Evidence
Q8	“Sshubam Verma ¹ Mohammed Safi Ur Rahman Khan ^{1,2} Vishwajeet Kumar ³ Rudra Murthy ³ Jaydeep Sen ³ 1Nilekani Centre at AI4Bharat 2Indian Institute of Technology, Madras 3IBM Research, India” (p.1)
Q22	“These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers.” (p.3)
Q86	“We would like to thank EkStep Foundation and Nilekani Philanthropies for their generous grant towards building datasets, models, tools and other resources for Indian languages.” (p.9)

Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MILU | Context: Bangladeshi Bengali University Student — Story Generation Thesis Deployment

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU evaluates models using accuracy on closed MCQ items via log-likelihood (non-API) or structured JSON parsing (API) [Q48, Q49, Q51, Q52, Q53]. The deployment requires evaluating open-ended Bengali story generation, which demands free-form text output assessed for narrative coherence, register appropriateness, cultural authenticity, and grounding in Bangladeshi named entities. No MILU mechanism produces a score that transfers to story generation evaluation [DATASET-D5 concern]. The authors themselves note log-likelihood may diverge from generation-based and CoT methods [Q85]. The output-form gap is structural: MCQ accuracy measures forced-choice factual recall, not generative cultural grounding. Even if MILU scores were perfectly calibrated to Bangladeshi knowledge (which they are not), they would still not validate story generation quality.

ID	Evidence
Q48	"For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations." (p.5)
Q49	"We use the log-likelihood method, where the probability of a given output string is computed by conditioning it on some provided input (Brown et al., 2020)." (p.5)
Q51	"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i))$ " (p.5)
Q52	"The API-based models are evaluated using the generative approach due to the lack of support for prompt log probabilities." (p.5)
Q53	"We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing." (p.5)
Q85	"Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting." (p.9)
D5	৬৫তম ফিল্মফেয়ার অ্যাওয়ার্ডস, ২০২০-এ সেরা সনিমোটোগ্রাফি জন্য পুরস্কারটিকে জিতছেন? ... জয় ওজা — 65th Filmfare Awards (Indian film industry awards) — not Bangladeshi film industry

Strengths

- MILU's evaluation infrastructure is well-documented and reproducible across 0/1/5-shot conditions, log-likelihood, and generative-JSON modes [Q46, Q48, Q53], spanning 42–45 models [Q44, Q75]. This breadth means baseline factual-recall comparisons across models are feasible as a screening step.
- Both base and instruct variants are evaluated [Q45], and per-subject/per-language tables are provided [Q99, Q128], enabling fine-grained baseline comparisons within MILU's intended construct.

ID	Evidence
Q44	"We evaluate 42 different models on MILU, including large proprietary models, open-source multilingual models, and popular fine-tuned models specific to Indic languages." (p.5)
Q45	"Both the base versions and instruction fine-tuned variants of these models, wherever applicable, are evaluated to measure the improvements gained from fine-tuning." (p.5)
Q46	"All models, except for proprietary models and LLAMA-3.1-405B, are tested under 0-shot, 1-shot, and 5-shot setups." (p.5)
Q48	"For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations." (p.5)
Q53	"We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing." (p.5)

Q75	"We evaluate 45 different LLMs and find that the majority of LLMs struggle on MILU, with GPT4o achieving the highest average accuracy." (p.8)
Q99	"Table 9: Detailed subject level statistics of MILU across different languages." (p.20)
Q128	"Table 47: Detailed subject-wise evaluation for GOOGLE/GEMMA-2-9B-IT on MILU across different languages." (p.52)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: No — MILU produces single discrete labels (option1–4) or structured JSON [Q51, Q53]; the deployment requires open-ended free-form Bengali narrative text. Output modalities do not match.

OF-2: Not applicable — deployment is text-only; no TTS requirement. INSUFFICIENT DOCUMENTATION on whether MILU's evaluation infrastructure could be extended to speech, but this is not a deployment requirement.

OF-3: Target population is highly literate (university students [WEB-3]); literacy is not a barrier for either MILU's or the deployment's text format.

OF-4: Severe external-validity violation: MCQ accuracy on India-centric content does not predict Bangladeshi story-generation quality. The deployment must construct a separate generative evaluation rubric (per BanglaSocialBench precedent [WEB-16]).

ID	Evidence
Q51	"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i \cdot))$ " (p.5)
Q53	"We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing." (p.5)
Q85	"Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting." (p.9)
WEB-3	University student literacy supports text-based deployment (https://en.wikipedia.org/wiki/Education_in_Bangladesh)
WEB-16	BanglaSocialBench's native-speaker-written, context-dependent evaluation methodology is the closest precedent for a story-generation rubric (https://arxiv.org/abs/2603.15949)

Information Gaps

- No story-generation evaluation methodology exists in MILU; supplementation is structural, not incremental

Requires Expert Verification

- Design and validation of a Bangladeshi story-generation rubric by native-speaker linguists and literary scholars

Remediation

Gap 1: MCQ accuracy via log-likelihood [Q51] does not transfer to evaluating open-ended Bengali narrative generation

Recommendation: Adopt a generation-based evaluation protocol (per the limitation noted in [Q85]) with multi-criteria human scoring (cultural authenticity, register, named-entity accuracy, narrative coherence) plus optional LLM-as-judge calibration validated against Bangladeshi annotator scores. Treat MILU accuracy purely as an upstream screening signal.

ID	Evidence
Q51	“For multiple choice questions, given k possible answer strings, we select the answer string (a _i) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i \cdot))$ ” (p.5)
Q85	“Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting.” (p.9)